

ASEN 6337 Group Project

The EuroSAT classification competition

How it runs, how it is scored, and how teams using different methods are compared on the same footing

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This one document is for everybody: for the students who are entering, for the professor who is overseeing it, and for me as the person running it. Section 6 is the one to read if you want to know how a team using principal components and clustering is judged against a team using a neural network.

Contents

1. What this is	2
2. What every team receives	2
3. What you send me.....	2
4. The dates	3
5. How a submission is scored.....	5
6. Teams using different methods, compared fairly	7
7. How the score becomes marks.....	10
8. Where the marks come from overall.....	10
9. The rules.....	11
10. Questions I expect.....	11

1. What this is

Every group in this class works on the same problem. You are given 18,900 small satellite images that are already labelled with what is on the ground, and 8,100 more that are not labelled. Your job is to work out the labels for those 8,100.

You build whatever model you like. You send me a file of your answers. I compare your answers with the correct ones, which only I hold, and I send back a score and a class leaderboard. There are two rounds of this, then a final submission.

The competition is worth 10 marks out of 100. The written report is worth 50 and the oral presentation 40. I want to be plain about that from the start: the leaderboard is there to give you a target and something to argue about. It is not the project. A team with a simple model and a careful report will finish above a team with a complicated model and a thin one.

2. What every team receives

Every team downloads exactly the same package. Nobody gets anything extra.

File	What it holds
train.npz or train.mat	18,900 images with their correct labels
test.npz or test.mat	8,100 images with no labels, shuffled, with random id numbers
sample_submission.csv	The exact file format to copy, with all 8,100 ids already filled in
baseline.ipynb	A simple model I wrote, which runs in about a minute, and its score
load_data.py and load_data.m	A few lines to load the data in Python or in MATLAB
README.md	The rules and the file format

Each image is 64 pixels by 64 pixels, and every pixel carries 13 numbers rather than the usual 3. Those 13 numbers are measurements of sunlight reflected back in 13 different wavelength ranges, from blue light through to infrared that the eye cannot see. A normal photograph gives you red, green and blue. This gives you ten more, and several of them are the ones that separate a field of one crop from a field of another.

There are 10 possible labels: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake.

Use the .npz files if you work in Python and the .mat files if you work in MATLAB. They hold identical data. Pick one and ignore the other.

3. What you send me

One file. A plain text file with two columns and 8,101 lines: a heading line, then one line for each of the 8,100 test images.

```
id,label
359356,AnnualCrop
752130,Forest
206827,SeaLake
...
```

The id is a number I attached to each test image when I built the package. You do not make these up. They are already inside the test file, and they are already written into `sample_submission.csv` for you. The easiest way to produce your submission is to take that sample file, leave the id column exactly as it is, and overwrite the label column with your own answers.

The id exists for one reason: so that I can line your answers up with mine even if your rows end up in a different order. If a row is missing, repeated, or carries a class name that is spelled wrong, my script refuses the file and tells you which row is at fault. Nothing is silently marked wrong.

Upload it to the Canvas assignment for that round. One upload per team, since the Canvas assignment is set up by group. That is the only thing you hand in during the competition. No code, no images, no probabilities.

Canvas records the time you uploaded, so there is never an argument about whether something arrived before the deadline. It also means you do not have to name the file in any particular way. Canvas already knows which team it came from.

4. The dates

When	What happens
Monday 5 October	I release the data package
Monday 2 November	Round 1 predictions due on Canvas
Wednesday 4 November	I email round 1 scores and the leaderboard
Monday 9 November	Round 2 predictions due on Canvas
Wednesday 11 November	I email round 2 scores and the leaderboard
Friday 13 November, 5 pm	Canvas closes. Your last upload is your final entry
Saturday 14 November	I score the final entries
Sunday 15 November	I email every team its final score
Monday 16 and Wednesday 18 November	Oral presentations
Friday 20 November	Written report and code due

You may upload as many times as you like before a deadline. Canvas keeps the most recent upload and that is the one I score, so treat it as replacing what came before. Uploading something in round 1, even from a rough model, is the cheapest way to find out that your file format is right before it matters.

5. How a submission is scored

I never run your code

I hold an answer key: a list of all 8,100 test ids with the correct label beside each one. I built it when I made the package and no student sees it. When your file arrives, one script reads your file and my key, matches the rows by id, and counts how many of your answers agree with mine. The same script runs on every team's file.

It works the way a multiple choice exam is marked. There are 8,100 questions and 10 possible answers to each. The marker does not care how you arrived at your answers. The marker lays your answer sheet next to the key and counts.

The score: macro-F1

The obvious score is accuracy, which is just the fraction you got right. I use a slightly different one called macro-F1, and it is worth understanding why.

For one class, say Forest, ask two questions. Of all the images you called Forest, how many really were Forest? That fraction is called precision. Of all the images that really were Forest, how many did you find? That fraction is called recall. F1 combines the two into a single number between 0 and 1, and it is only high when both are high. Work that out for each of the 10 classes and take the plain average. That average is macro-F1.

A small example with ten images and three classes, F for Forest, R for River, H for Highway:

	1	2	3	4	5	6	7	8	9	10
Correct answer	F	F	F	F	R	R	R	H	H	H
Your answer	F	F	F	R	F	R	R	H	H	F

Seven answers out of ten are right, so accuracy is 0.70. Now go class by class. You called five images Forest, which were images 1, 2, 3, 5 and 10, and three of those five really are Forest, so precision for Forest is 3 divided by 5, which is 0.60. There are four real Forest images and you found three of them, so recall is 3 divided by 4, which is 0.75. Combining those gives an F1 of 0.67 for Forest. The same arithmetic gives River 0.67 and Highway 0.80. Macro-F1 is the average of the three, which is 0.71.

The reason I use it is that the classes are not all the same size, and accuracy lets a model hide a failure. A model that handles nine classes perfectly and never once predicts Pasture would still score about 0.93 accuracy, because Pasture is only a small part of the test set. Its macro-F1 would be at most 0.90, because one of the ten class scores would be zero. Macro-F1 makes you pay for every class you give up on, and in this dataset the classes people give up on are the interesting ones.

Along with your score I send back a table showing precision, recall and F1 for each of the 10 classes separately. That table is more useful to you than the single number, because it tells you where to spend your next week.

6. Teams using different methods, compared fairly

This is the part people worry about, and it deserves a careful answer. Different groups will take different routes. One team might reduce the data with principal components and then cluster it. Another might use principal components and then a neural network. A third might ignore the images and work from simple per-band statistics with a random forest. How can those be compared without favouring one?

6.1 The comparison never looks at your method

The answer is that the comparison does not happen between the methods at all. It happens between the answer files. Whatever you build, the thing that leaves your computer is the same two-column file: an id and a class name. By the time I see your work, every trace of how you produced it is gone.

Team	What they do inside	What comes out
A	Principal components, then k-means clustering	id and class name for 8,100 images
B	Principal components, then a support vector machine	id and class name for 8,100 images
C	Per-band averages, then a random forest	id and class name for 8,100 images
D	A convolutional network on the raw images	id and class name for 8,100 images

Four different routes, one shape of answer. I open four files that look the same, run the same script on each, and compare each against the same key. There is no setting in my script for "this one is a neural network".

6.2 What is identical for everyone

- The same 18,900 labelled images to learn from.
- The same 8,100 test images to label.
- The same answer key, written once before the competition opened and never changed.
- The same scoring script, and the same measure, macro-F1.
- The same deadlines, and the same rule that the most recent upload is the one scored.
- The same rule that everything is trained from scratch on the released data, with no pretrained weights and no outside data.

6.3 What you are free to choose

- Which of the 13 bands you use, and whether you build combinations of them.
- Whether you work from whole images or from summary numbers per image.
- Whether you reduce dimensions first, with principal components or anything else.
- Which model you fit, and how you set it up.

- How you split off your own validation set from the training data.
- Whether you augment the training images, and how.

None of these choices earns or loses marks by itself. They earn marks through the reasons you give for them in your report and through the score they produce.

6.4 A note for teams using clustering

Clustering deserves its own paragraph, because it does not produce class names on its own. If you run k-means you get groups numbered 1, 2, 3 and so on, and nothing tells you that group 3 means Forest. You still have to hand me one of the ten class names for each test image.

The usual way to bridge that gap is to cluster first and then look at the training labels inside each cluster. Whichever real class is most common in a cluster becomes the name you give to everything in it. That is allowed, and it is a sensible thing to do. It also makes an interesting report: you can say how clean the clusters were, which two classes kept landing in the same cluster, and what that tells you about the data. A team that does this well and scores slightly lower than a neural network can still finish ahead overall.

6.5 Why the single number is comparable

A score is comparable across methods when it is produced by the same measuring process on the same material. All 8,100 test images are judged for every team. No team is asked an easier set of questions. Macro-F1 weighs each of the ten classes the same way for everyone, so no method can win by quietly abandoning a class that another method attempted.

There is one honest imperfection and I would rather state it than hide it. I split the test images into two halves at random. Because the split is random, a score carries a small amount of luck, roughly four tenths of a percentage point either way. That is why marks are awarded in bands rather than by ranking teams against each other. Two teams half a point apart are not meaningfully different, and the marking does not pretend otherwise.

6.6 The practice half and the final half

Those two halves work like a practice paper and a final paper drawn from the same set of questions. During the competition the leaderboard shows your score on the practice half only. After the close, your last submission is scored on the final half, and that is the score that counts for marks.

The reason is to stop a team improving its mark by sending many slightly different files and keeping whichever happened to score well. That is not learning anything about the data, it is learning about the particular images in the leaderboard. If your final score comes out well below your practice score, that is what happened, and I will ask you about it in the oral.

6.7 The larger point

Ninety of the hundred marks come from the report and the oral, where I am reading your reasoning rather than your score. Why you chose your method. How you set up your own

validation. Which classes your model confuses and what it is about those places on the ground that explains it. What you tried that did not work.

A team using principal components and clustering can score full marks on all of that. A team using a neural network can lose most of it. The competition decides 10 marks. Your thinking decides the other 90.

7. How the score becomes marks

The 10 marks are split as follows.

Two marks for sending a valid final entry: on time, in the right format, with an answer for every test image.

Eight marks for the score itself, measured against two fixed numbers that I publish when the competition opens. The first is my baseline, called B, which is a simple model using nothing but per-band averages. The second is a target, called T, from a model trained from scratch without anything clever. Matching the baseline earns 4. Reaching the target earns 8. Anything in between earns marks in proportion. Below the baseline earns 0.

Your final macro-F1	Marks	Why
Below B	0	Below the simple model you were given
Equal to B	4	Matched the simple model
A quarter of the way from B to T	5	
Halfway from B to T	6	
Three quarters of the way	7	
Equal to T or above	8	Reached the target

There are no marks for finishing first, second or third. I will announce the top three in class and that is the whole of the reward. This is deliberate. Because everybody is measured against the same two fixed numbers, your marks do not depend on how well anyone else did, and every team in the class can earn all 10.

8. Where the marks come from overall

Component	Marks	What it is for
Written report	50	Your question, your method and the reasons for it, your experiments, your error analysis, your limitations
Oral presentation	40	Explaining the work and answering questions on it
Competition	10	A valid final entry and the score it earned
Total	100	

The largest single item inside the report is the error analysis. Which classes your model confuses, what a few of its mistakes look like when you put the images on the page, and what it is about those places on the ground that explains the confusion. A confusion matrix you discuss properly is worth more than a clever diagram you do not.

9. The rules

1. Train from scratch on the released training data. No pretrained weights, and no data from outside the package.
2. Do not try to find the test labels. EuroSAT is a public dataset, so in principle the test images could be traced back to a public copy. Doing that is a breach of the honour code, and with 10 marks at stake it is a poor trade.
3. Work within your own team. Anything you post publicly to the whole class is fine and is available to everybody. Quietly sharing code or predictions between two teams is not.
4. Upload to Canvas by the deadline for each round. You may replace your upload as often as you like before it closes.
5. Hand in your code with the report. I use it to judge whether someone else could repeat your work, not to produce your score. I do not re-run it.
6. Declare any use of generative AI as the course policy requires. I will never treat writing style as evidence of anything. If something in a report needs checking, I ask about it in the oral.

10. Questions I expect

Can I use a method we have not covered in class?

Yes. You have to be able to explain it and defend the choice, which in practice is the real limit.

Is a neural network expected?

No. The baseline I give you uses no spatial information at all and still does reasonably well. There is room above it for several kinds of model.

We used clustering and our score is lower than our neighbours. Have we lost?

Not at all. The gap is worth a mark or two out of a hundred. Your report and your oral are worth ninety.

What if two teams end up with the same score?

Nothing happens. Marks come from the bands in section 7, not from ranking, so an equal score means equal marks.

Can we use all 13 bands, or should we pick a few?

Either. Working out how few you can get away with, and showing the evidence, is a perfectly good project in itself.

What if our file is rejected?

I tell you what is wrong with it. If the round is still open, upload a corrected one to Canvas, which replaces the broken file. If the round has closed, you carry no score for that round, which is why it is worth uploading early.